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Read synchronization delay at subordinate device for 1 wire protocol

Abstract

A subordinate device includes a local clock and a protocol controller. The protocol controller is coupled to a bus that includes a data wire. The protocol controller is configured to receive, on the data wire from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction. The protocol controller is also configured to, after a delay window, send synchronization pulses to the principal device based on the local clock, where a duration of the delay window is greater than a clock cycle of the local clock. The protocol controller is further configured to, after sending the synchronization pulses, send data on the data wire to the principal device based on the local clock during the read transaction.

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Background/Summary**I. FIELD**

(1) The present disclosure is generally related to read synchronization delay at subordinate device for 1 wire protocol.

II. DESCRIPTION OF RELATED ART

(2) Advances in technology have resulted in smaller and more powerful computing devices. For example, there currently exist a variety of portable personal computing devices, including wireless telephones such as mobile and smart phones, tablets and laptop computers that are small, lightweight, and easily carried by users. These devices can communicate voice and data packets over wireless networks. Further, many such devices incorporate additional functionality such as a digital still camera, a digital video camera, a digital recorder, and an audio file player. Also, such devices can process executable instructions, including software applications, such as a web browser application, that can be used to access the Internet. As such, these devices can include significant computing capabilities.

(3) Such computing devices often include components, such as processing circuits, antennas, filters, storage devices, peripheral devices, etc., that communicate via a serial bus. The MIPI® (a

registered trademark of MIPI Alliance, Inc.) alliance has defined a Radio Frequency Front End (RFFE) interface for communication among radio frequency (RF) front-end devices. For example, the RFFE interface enables communication between a principal device and one or more subordinate devices (e.g., RFFE components) over a serial bus. The principal device may also be referred to as a “master” device and a subordinate device may also be referred to as a “slave” device.

(4) According to the RFFE interface, in a two-wire protocol, a read command from the principal device over the serial bus to a subordinate device is followed by a bus park after which the subordinate device is expected to provide the requested data on the serial bus. Typically, the bus park has a one clock cycle duration that can be insufficient for the requested data to be available to send at the subordinate device. The data retrieved by the principal device from the serial bus subsequent to the bus park can thus be invalid.

(5) Similarly, in a one-wire protocol, a read command from the principal device over the serial bus to a subordinate device is followed by a bus park after which the subordinate device is expected to send synchronization pulses and then provide the requested data on the serial bus. Typically, the bus park has a one clock cycle duration that can be insufficient for the requested data to be available to send at the subordinate device. The data retrieved by the principal device from the serial bus subsequent to the synchronization pulses can thus be invalid.

III. SUMMARY

(6) According to one implementation of the present disclosure, a subordinate device includes a local clock and a protocol controller. The protocol controller is coupled to a bus that includes a data wire. The protocol controller is configured to receive, on the data wire from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction. The protocol controller is also configured to, after a delay window, send synchronization pulses to the principal device based on the local clock, where a duration of the delay window is greater than a clock cycle of the local clock. The protocol controller is further configured to, after sending the synchronization pulses, send data on the data wire to the principal device based on the local clock during the read transaction.

(7) According to another implementation of the present disclosure, a method includes receiving, at a subordinate device on a data wire of a bus from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction. The method also includes, after a delay window, sending synchronization pulses from the subordinate device to the principal device based on a local clock of the subordinate device, where a duration of the delay window is greater than a clock cycle of the local clock. The method further includes, after sending the synchronization pulses, sending data from the subordinate device on the data wire to the principal device based on the local clock during the read transaction.

(8) According to another implementation of the present disclosure, a non-transitory computer-readable medium stores instructions that, when executed by one or more processors, cause the one or more processors to receive, at a subordinate device on a data wire of a bus from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction. The instructions also cause the one or more processors to, after a delay window, send synchronization pulses from the subordinate device to the principal device based on a local clock of the subordinate device, where a duration of the delay window is greater than a clock cycle of the local clock. The instructions further cause the one or more processors to, after sending the synchronization pulses, send data from the subordinate device on the data wire to the principal device based on the local clock during the read transaction.

(9) Other aspects, advantages, and features of the present disclosure will become apparent after review of the entire application, including the following sections: Brief Description of the Drawings, Detailed Description, and the Claims.

Description

IV. BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram of a particular illustrative aspect of a system operable to configure a read synchronization delay at a subordinate device for 1 wire protocol, in accordance with some examples of the present disclosure.

(2) FIG. 2A is a diagram of an illustrative aspect of operations associated with a particular configuration of a read synchronization delay at a subordinate device for 1 wire protocol, in accordance with some examples of the present disclosure.

(3) FIG. 2B is a diagram of an illustrative aspect of operations associated with another particular configuration of a read synchronization delay at a subordinate device for 1 wire protocol, in accordance with some examples of the present disclosure.

(4) FIG. 2C is a diagram of an illustrative aspect of operations associated with another particular configuration of a read synchronization delay at a subordinate device for 1 wire protocol, in accordance with some examples of the present disclosure.

(5) FIG. 3 is a diagram of a particular implementation of a method of configuration of a read synchronization delay at a subordinate device for 1 wire protocol that may be performed by the system of FIG. 1, in accordance with some examples of the present disclosure.

(6) FIG. 4 illustrates various electronic devices that may integrate an electronic circuit, and/or an integrated device, described herein.

V. DETAILED DESCRIPTION

(7) In a one-wire protocol, a read command from a principal device over a serial bus to a subordinate device is followed by a bus park after which the subordinate device is expected to send synchronization pulses and then provide the requested data on the serial bus. Typically, the bus park has a one clock cycle duration that can be insufficient for the requested data to be available to send at the subordinate device. The data retrieved by the principal device from the serial bus subsequent to the synchronization pulses can thus be invalid.

(8) To retrieve valid data from the serial bus, the principal device can send multiple identical read requests. For example, the principal device can send a first read request on the serial bus, ignore data on the serial bus after receiving synchronization pulses after a one clock cycle bus park subsequent to the first read request, send a second read request on the serial bus, and read data on the serial bus after receiving synchronization pulses after a one clock cycle bus park subsequent to the second read request. When the one clock cycle bus park subsequent to the second read request is complete, the subordinate device is more likely to have the requested data available to send and the principal device is more likely to retrieve valid data from the serial bus. However, sending multiple requests for each read reduces data throughput and increases latency.

(9) Systems and methods of configuring a read synchronization delay at a subordinate device for 1 wire protocol are disclosed. As an example, a principal device is coupled via a serial bus to one or more subordinate devices. The principal device sends, via a data wire of the serial bus, a read command addressed to a subordinate device to initiate a read transaction. The data wire is put in a bus park subsequent to sending of the read command. In some examples, the principal device puts the data wire in a bus park subsequent to sending the read command. In some examples, the subordinate device puts the data wire in a bus park subsequent to receiving the read command.

(10) If the subordinate device determines at the end of the bus park that the requested data is available to send, the subordinate device sends synchronization pulses on the data wire after the bus park and then sends the data to complete the read transaction. On the other hand, if the subordinate device determines that the requested data is not available to send at the end of the bus park, the subordinate device keeps the data wire in an extended bus park during an added delay for up to a delay threshold. The subordinate device, upon arrival of the data during the added delay, sends the

synchronization pulses, and sends the data to complete the read transaction. Alternatively, the subordinate device, in response to determining that the added delay has reached the delay threshold prior to arrival of the data, sends termination pulses to terminate the read transaction. The added delay increases a likelihood that the requested data arrives at the subordinate device to send and increases a likelihood that the principal device retrieves valid data after the added delay. Selectively sending the synchronization pulses after an added delay has a lower impact on data throughput and latency compared to sending multiple requests for each read.

(11) Particular aspects of the present disclosure are described below with reference to the drawings. In the description, common features are designated by common reference numbers. As used herein, various terminology is used for the purpose of describing particular implementations only and is not intended to be limiting of implementations. For example, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Further, some features described herein are singular in some implementations and plural in other implementations. To illustrate, FIG. 1 depicts a device **102** including one or more subordinate devices (“subordinate device(s)” **122** of FIG. 1), which indicates that in some implementations the device **102** includes a single subordinate device **122** and in other implementations the device **102** includes multiple subordinate devices **122**. For ease of reference herein, such features are generally introduced as “one or more” features and are subsequently referred to in the singular or optional plural (as indicated by “(s)”) unless aspects related to multiple of the features are being described.

(12) In some drawings, multiple instances of a particular type of feature are used. Although these features are physically and/or logically distinct, the same reference number is used for each, and the different instances are distinguished by addition of a letter to the reference number. When the features as a group or a type are referred to herein e.g., when no particular one of the features is being referenced, the reference number is used without a distinguishing letter. However, when one particular feature of multiple features of the same type is referred to herein, the reference number is used with the distinguishing letter. For example, referring to FIG. 1, multiple subordinate devices are illustrated and associated with reference numbers **122A** and **122B**. When referring to a particular one of these subordinate devices, such as a subordinate device **122A**, the distinguishing letter “A” is used. However, when referring to any arbitrary one of these subordinate devices or to these subordinate devices as a group, the reference number **122** is used without a distinguishing letter.

(13) As used herein, the terms “comprise,” “comprises,” and “comprising” may be used interchangeably with “include,” “includes,” or “including.” Additionally, the term “wherein” may be used interchangeably with “where.” As used herein, “exemplary” indicates an example, an implementation, and/or an aspect, and should not be construed as limiting or as indicating a preference or a preferred implementation. As used herein, an ordinal term (e.g., “first,” “second,” “third,” etc.) used to modify an element, such as a structure, a component, an operation, etc., does not by itself indicate any priority or order of the element with respect to another element, but rather merely distinguishes the element from another element having a same name (but for use of the ordinal term). As used herein, the term “set” refers to one or more of a particular element, and the term “plurality” refers to multiple (e.g., two or more) of a particular element.

(14) As used herein, “coupled” may include “communicatively coupled,” “electrically coupled,” or “physically coupled,” and may also (or alternatively) include any combinations thereof. Two devices (or components) may be coupled (e.g., communicatively coupled, electrically coupled, or physically coupled) directly or indirectly via one or more other devices, components, wires, buses, networks (e.g., a wired network, a wireless network, or a combination thereof), etc. Two devices (or components) that are electrically coupled may be included in the same device or in different devices and may be connected via electronics, one or more connectors, or inductive coupling, as illustrative, non-limiting examples. In some implementations, two devices (or components) that are

communicatively coupled, such as in electrical communication, may send and receive signals (e.g., digital signals or analog signals) directly or indirectly, via one or more wires, buses, networks, etc. As used herein, “directly coupled” may include two devices that are coupled (e.g., communicatively coupled, electrically coupled, or physically coupled) without intervening components.

(15) In the present disclosure, terms such as “determining,” “calculating,” “estimating,” “shifting,” “adjusting,” etc. may be used to describe how one or more operations are performed. It should be noted that such terms are not to be construed as limiting and other techniques may be utilized to perform similar operations. Additionally, as referred to herein, “generating,” “calculating,” “estimating,” “using,” “selecting,” “accessing,” and “determining” may be used interchangeably. For example, “generating,” “calculating,” “estimating,” or “determining” a parameter (or a signal) may refer to actively generating, estimating, calculating, or determining the parameter (or the signal) or may refer to using, selecting, or accessing the parameter (or signal) that is already generated, such as by another component or device.

(16) Referring to FIG. 1, a particular illustrative aspect of a system **100** operable to configure a read synchronization delay at a subordinate device for 1 wire protocol is disclosed. The system **100** includes a device **102** that includes a principal device **120** coupled via a serial bus **130** to one or more subordinate devices **122**, such as a subordinate device **122A**, a subordinate device **122B**, one or more additional subordinate devices, or a combination thereof.

(17) The principal device **120** includes a protocol controller **140** and a clock **144** (e.g., a local clock). A subordinate device **122** includes a protocol controller **142** and a clock **146** (e.g., a local clock). For example, the subordinate device **122A** includes a protocol controller **142A** and a local clock **146A**. As another example, the subordinate device **122B** includes a protocol controller **142B** and a local clock **146B**. The serial bus **130** includes a data wire **134**. The serial bus **130** corresponds to a “one-wire” bus. For example, the serial bus **130** does not include a clock wire. The serial bus **130** may include a ground wire (not shown) in addition to the data wire **134**.

(18) In some implementations, the protocol controller **140** of the principal device **120** is integrated in an RF front-end device. In some aspects, the principal device **120** and the one or more subordinate devices **122** operate in compliance with a one-wire protocol. In a particular aspect, the principal device **120** and a subordinate device **122** correspond to a “master” device and a “slave” device, respectively, in compliance with the one-wire protocol. In a particular aspect, the principal device **120** (e.g., the protocol controller **140**) is configured to initiate communication on the data wire **134**. For example, the principal device **120** (e.g., the protocol controller **140**) is configured to send one or more commands on the data wire **134** to the one or more subordinate devices **122**. In a particular aspect, the subordinate device **122** (e.g., the protocol controller **142**) is configured to perform one or more commands received on the data wire **134**. As illustrative, non-limiting examples, a command can include a register read command to read data from a particular register, or a register write command to write data to a particular register.

(19) In some implementations, components of the device **102** can switch roles between principal or subordinate device, with a single component coupled to the serial bus **130** corresponding to a principal device **120** at a given time. To illustrate, for a first transaction, a particular component coupled to the serial bus **130** corresponds to a principal device **120** and the remaining components coupled to the serial bus **130** correspond to subordinate devices **122**. For a second transaction, another component coupled to the serial bus **130** can correspond to a principal device **120** and the remaining components coupled to the serial bus **130** can correspond to subordinate devices **122**.

(20) In some implementations, the principal device **120** includes a baseband processor of the device **102**. In some implementations, the subordinate device(s) **122** include RFFE components, such as a power amplifier, a transmitter, a filter, a switch, a receiver, a low-noise amplifier, an antenna, a tuner, a processing circuit, a storage device, or a combination thereof. Although the principal device **120** and the subordinate device(s) **122** are illustrated as internal to the device **102**, in other

implementations the principal device **120**, the subordinate device(s) **122**, or a combination thereof, can be external to the device **102** and coupled to the serial bus **130**.

(21) During operation, the protocol controller **142** obtains data indicating a delay threshold **148** to be used by the protocol controller **142** during read transactions with the principal device **120**. For example, the protocol controller **142A** obtains data indicating a delay threshold **148A**. As another example, the protocol controller **142B** obtains data indicating a delay threshold **148B**.

(22) In a particular aspect, the delay threshold **148** is based on memory access latency, a configuration of the serial bus **130**, or both, associated with a subordinate device **122**. For example, the delay threshold **148A** is based on a memory access latency associated with data reads performed by the protocol controller **142A** of the subordinate device **122A**. To illustrate, the memory access latency corresponds to an expected delay between the protocol controller **142A** initiating a read at a register and obtaining data from the register at the subordinate device **122A**.

(23) As another example, the delay threshold **148A** is based on a configuration of the data wire **134** between the principal device **120** and the subordinate device **122A**. The configuration of the data wire **134** can include a line length **150** associated with the subordinate device **122A** that corresponds to an expected delay between sending data from the subordinate device **122A** and the data being received at the principal device **120**. To illustrate, the line length **150** is based on a length **152** of a first wire between the principal device **120** and a connection point (CP) **162** of the data wire **134**, a length **154** of the data wire **134** between the connection point **162** and a connection point **164** of the data wire **134**, and a length **156** of a second wire between the connection point **164** and the subordinate device **122A**. The data wire **134** between the connection point **162** and the connection point **164** corresponds to a portion of the shared bus (e.g., the serial bus **130**). Similarly, in a particular aspect, the delay threshold **148B** is based on at least one of memory access latency, a configuration of the serial bus **130**, or both, associated with the subordinate device **122B**.

(24) In some aspects, the delay threshold **148A** is distinct from the delay threshold **148B**. In an example, the delay threshold **148A** is less than the delay threshold **148B** because at least one of the line length **150** or memory access latency of the subordinate device **122A** is less than at least a corresponding one of a line length or memory access latency of the subordinate device **122B**. To illustrate, a duration of the delay threshold **148B** is longer than a duration of the delay threshold **148A**.

(25) In some aspects, the delay threshold **148A** is the same as the delay threshold **148B**. For example, the delay threshold **148A** is the same as the delay threshold **148B** because at least one of the line length **150** or memory access latency of the subordinate device **122A** is equal to at least one of a line length or memory access latency, respectively, of the subordinate device **122B**. In another example, the delay threshold **148A** is the same as the delay threshold **148B** because the delay threshold **148A** and the delay threshold **148B** are based on a representative (e.g., average) delay threshold. In a particular aspect, the data indicating one or more delay thresholds **148** is based at least in part on a configuration setting, default data, a user input, or a combination thereof. In a particular aspect, the data indicating one or more delay thresholds **148** is based at least in part on dynamically detected memory access latency, dynamically detected configuration of the serial bus **130**, or both, associated with the subordinate device(s) **122**.

(26) The protocol controller **140** of the principal device **120** sends, on the data wire **134** of the serial bus **130**, a read command addressed to a subordinate device **122A** to initiate a read transaction, as further described with reference to FIG. 2A. For example, the read command identifies a register from which the subordinate device **122A** is to read data to send on the data wire **134**. In a particular aspect, the protocol controller **140** sends the read command in response to receiving a data request from another component of the device **102**.

(27) The data wire **134** is put into a bus park when data on the data wire **134** switches direction. In some implementations, the protocol controller **140** of the principal device **120** puts the data wire **134** into a bus park after sending the read command on the data wire **134**. In alternative

implementations, the protocol controller **142A** of the subordinate device **122A** puts the data wire **134** in a bus park responsive to receiving the read command addressed to the subordinate device **122A**.

(28) The protocol controller **142A** of the subordinate device **122A**, in response to receiving the read command on the data wire **134** addressed to the subordinate device **122A**, initiates a read of the requested data from the register indicated in the read command, as further described with reference to FIG. 2A. The protocol controller **142A**, in response to arrival of the data at the subordinate device **122A** prior to an end of the bus park, sends one or more synchronization pulses on the data wire **134** and then sends the data on the data wire **134**, as further described with reference to FIG. 2A. In this example, the synchronization pulses are sent subsequent to a delay window corresponding to a duration (e.g., one clock cycle) of the bus park. The protocol controller **140** of the principal device **120**, in response to detecting the synchronization pulses on the data wire **134**, reads the data on the data wire **134** and the read transaction is completed successfully.

(29) Alternatively, the protocol controller **142A** of the subordinate device **122A**, in response to detecting an end of the bus park prior to arrival of the requested data at the subordinate device **122A**, adds a delay that is based on the delay threshold **148A**. For example, the delay threshold **148A** indicates a particular count of clock cycles of the clock **146A**. The protocol controller **142**, in response to detecting arrival of the data at the subordinate device **122A** prior to elapse of the particular count of clock cycles of the clock **146A**, sends one or more synchronization pulses on the data wire **134** and then sends the data on the data wire **134**, as further described with reference to FIG. 2B. In this example, the synchronization pulses are sent subsequent to a delay window corresponding to a duration (e.g., one clock cycle) of the bus park and the added delay. The protocol controller **140** of the principal device **120**, after detecting the synchronization pulses on the data wire **134**, reads the data on the data wire **134** and the read transaction is completed successfully. The clock cycles of the clock **146A** that are added to the duration of the bus park prior to sending the one or more synchronization pulses on the data wire **134** correspond to a delay that is added by the subordinate device **122A** to effectively extend the bus park, as further described with reference to FIG. 2B. A technical advantage of adding the delay includes enabling the protocol controller **140** of the principal device **120** to retrieve valid data from the data wire **134**. A technical advantage of dynamically extending the delay window includes enabling the subordinate device **122A** to limit the latency that is added due to the delay window.

(30) In a particular aspect, the protocol controller **142A** of the subordinate device **122A**, in response to detecting that an added delay corresponding to the particular count of clock cycles indicated by the delay threshold **148A** has elapsed prior to arrival of the data at the subordinate device **122A**, sends one or more termination pulses on the data wire **134** to terminate the read transaction, as further described with reference to FIG. 2C. The protocol controller **140** of the principal device **120**, based on detecting the termination pulses on the data wire **134**, determines that the read transaction is unsuccessful and that the data wire **134** is available for sending any further commands to the one or more subordinate devices **122**. A technical advantage of extending the delay window up to the duration indicated by the delay threshold **148A** includes limiting a maximum latency that is added to the data wire **134** for a read transaction.

(31) The system **100** thus enables the subordinate device **122A** to have more time (when the bus park is effectively extended by an added delay) to retrieve the requested data. A technical advantage of using individual delay thresholds includes increasing a likelihood that a subordinate device **122** that has a higher delay threshold **148** (e.g., corresponding to higher memory latency or line length) has valid data at the end of a longer added delay.

(32) FIG. 2A is a diagram of an illustrative aspect of operations **200** associated with a particular configuration of a bus park delay, in accordance with some examples of the present disclosure.

(33) The protocol controller **140** of the principal device **120** initiates a read transaction. For example, the protocol controller **140** sends a read command on the data wire **134** in accordance

with clock cycles of the clock **144** (e.g., a local clock). To illustrate, the protocol controller **140** sends, on the data wire **134**, a sequence start condition (SSC) **210** followed by a register read command frame **202**. The register read command frame **202** indicates a subordinate device address **212** of a subordinate device **122** and a command identifier **214** (e.g., **011**) of a register read command. The register read command frame **202** also includes a register identifier **216** of a register **226** from which the subordinate device **122** is to read data. The register read command frame **202** may also include a parity bit **218**. The register **226** is coupled to the device **102** of FIG. **1**. In some examples, the register **226** is integrated in the device **102**. In other examples, the register **226** is external to the device **102**.

(34) The data wire **134** is placed in a bus park **220** subsequent to the register read command frame **202**. In a particular example, the protocol controller **140** places the data wire **134** in the bus park **220** subsequent to sending the register read command frame **202**. In an alternative example, the protocol controller **142** of the subordinate device **122**, responsive to determining that the register read command frame **202** indicates the subordinate device address **212** of the subordinate device **122**, places the data wire **134** in the bus park **220**.

(35) The protocol controller **142** of the subordinate device **122** receives, on the data wire **134** from the principal device **120**, a read command addressed to the subordinate device **122** indicating initiation of the read transaction. For example, the protocol controller **142** receives the register read command frame **202** indicating the subordinate device address **212** of the subordinate device **122**.

(36) The protocol controller **142** of the subordinate device **122** initiates a memory access of the requested data responsive to receiving the read command. For example, the protocol controller **142**, in response to receiving the register read command frame **202** indicating the subordinate device address **212** of the subordinate device **122**, sends a read request **280** to the register **226** indicated by the register identifier **216**.

(37) In the example illustrated in FIG. **2A**, the register **226**, responsive to the read request **280**, sends data **282** to the subordinate device **122**. The protocol controller **142** of the subordinate device **122**, in response to arrival of the data **282** prior to end of the bus park **220**, sends synchronization pulses **260** on the data wire **134** based on clock cycles of the clock **146** (e.g., a local clock). The synchronization pulses **260** are sent responsive to detecting an end of a duration (e.g., one clock cycle of the clock **146**) of the bus park **220**. The synchronization pulses **260** including two pulses is provided as an illustrative example, in other examples the synchronization pulses **260** can include a single pulse or more than two pulses.

(38) After sending the synchronization pulses **260**, the protocol controller **142** sends a data frame **204** corresponding to the data **282** on the data wire **134** based on clock cycles of the clock **146** during the read transaction. The protocol controller **140**, in response to detecting the synchronization pulses **260** on the data wire **134**, retrieves the data frame **204** representing the data **282** from the data wire **134** subsequent to the synchronization pulses **260** to end the read transaction. The principal device **120** thus receives valid data from the data wire **134** without the subordinate device **122** adding a delay subsequent to the bus park **220**.

(39) In some aspects, the data frame **204** is followed by a bus park **222**. For example, the protocol controller **142** places the data wire **134** in the bus park **222** subsequent to sending the data frame **204**. In another example, the protocol controller **140** of the principal device **120** places the data wire **134** in the bus park **222** subsequent to receiving the data frame **204**.

(40) FIG. **2B** is a diagram of an illustrative aspect of operations **290** associated with another particular configuration of a bus park delay, in accordance with some examples of the present disclosure.

(41) Responsive to the subordinate device **122** receiving the register read command frame **202** on the data wire **134** from the principal device **120** and concurrently with sending the read request **280** to the register **226** indicated by the register identifier **216**, as described with reference to FIG. **2A**, the protocol controller **142** initializes a counter **246** based on the delay threshold **148** (e.g., a

duration of added delay) and updates the counter **246** at each clock cycle of the clock **146** (e.g., a local clock). In some implementations, the protocol controller **142** initializes the counter **246** responsive to determining that data has not been received at the subordinate device **122** from the register **226** prior to an end of the bus park **220**. In these implementations, the protocol controller **142** uses the counter **246** to track a duration of an added delay **262** corresponding to a delay window **228**.

(42) The protocol controller **142**, subsequent to detecting an end of the bus park **220**, keeps the data wire **134** in a bus park during an added delay **262** while the counter **246** indicates that an end of the delay threshold **148** has not been reached. For example, the protocol controller **142**, in response to determining that data has not been received from the register **226** prior to the end of the bus park **220**, initializes the counter **246** to indicate a remaining count of clock cycles corresponding to the delay threshold **148**, and updates (e.g., decreases) the counter **246** (e.g., by 1) at each clock cycle of the clock **146**. The register **226**, in response to the read request **280**, provides data **282** to the subordinate device **122**. The protocol controller **142**, in response to detecting arrival of the data **282** at the subordinate device **122** and determining that the counter **246** indicates at least one remaining count of clock cycles, determines that the data **282** has arrived during the delay window **228**. The protocol controller **142**, in response to determining that the data **282** has arrived during the delay window **228**, sends the synchronization pulses **260** on the data wire **134** based on clock cycles of the clock **146** (e.g., a local clock). Because the synchronization pulses **260** are sent upon detecting arrival of the data **282**, a duration of the added delay **262** (and hence a duration of the delay window **228**) is based on a memory access time associated with arrival of the data **282** at the subordinate device **122**.

(43) After sending the synchronization pulses **260**, the protocol controller **142** sends a data frame **204** corresponding to the data **282** on the data wire **134** based on clock cycles of the clock **146** during the read transaction, as described with reference to FIG. 2A. The protocol controller **140**, in response to detecting the synchronization pulses **260** on the data wire **134**, retrieves the data frame **204** representing the data **282** from the data wire **134** subsequent to the synchronization pulses **260** to end the read transaction.

(44) A duration of the delay window **228** (e.g., including the bus park **220** and the added delay **264**) is longer than a duration of the bus park **220** (e.g., a clock cycle of the clock **146**). The delay window **228** thus enables the principal device **120** to retrieve valid data from the data wire **134** that may be unavailable to send from the subordinate device **122** at the end of the bus park **220**. A technical advantage of the added delay **262** can include increased throughput and reduced latency as compared to sending multiple requests for the same read.

(45) Although the protocol controller **142** is described as initializing the counter **246** in response to detecting that data has not arrived at the subordinate device **122** prior to an end of the bus park **220**, in other implementations the protocol controller **142** can initialize the counter **246** responsive to receiving the register read command frame **202** independently of detecting an end of the bus park **220**. For example, the protocol controller **142** can use the counter **246** to track a duration of the bus park **220** and, if data is not received prior to the end of the bus park **220**, also to track a duration of the added delay **262**. To illustrate, the protocol controller **142** initializes the counter **246** based on a sum of a duration of the bus park **220** (e.g., 1 clock cycle) and the delay threshold **148** and sends the synchronization pulses **260** responsive to detecting arrival of the data **282** at the subordinate device **122** when the counter **246** indicates at least one remaining clock cycle.

(46) FIG. 2C is a diagram of an illustrative aspect of operations **292** associated with another particular configuration of a bus park delay, in accordance with some examples of the present disclosure.

(47) Responsive to the subordinate device **122** receiving the register read command frame **202** on the data wire **134** from the principal device **120** and concurrently with sending the read request **280** to the register **226** indicated by the register identifier **216**, the protocol controller **142** initializes a

counter **246** based on the delay threshold **148** (e.g., a duration threshold of a delay window) and updates the counter **246** at each clock cycle of the clock **146** (e.g., a local clock), as described with reference to FIG. 2B.

(48) The protocol controller **142**, in response to detecting that the counter **246** has reached a value (e.g., 0) indicating no remaining clock cycles and that data from the register **226** has not arrived at the subordinate device **122**, determines that a delay window **230** has ended prior to arrival of data from the register **226** and sends termination pulses **270** on the data wire **134** based on a clock cycle of a local clock (e.g., the clock **146**). The delay window **230** includes the bus park **220** and an added delay **264** during which the data wire **134** effectively remains in bus park.

(49) The protocol controller **142** sends the termination pulses **270** to terminate the read transaction. In a particular aspect, the protocol controller **142** places the data wire **134** in a bus park **232** subsequent to sending the termination pulses **270**. The termination pulses **270** including two pulses is provided as an illustrative example, in other examples the termination pulses **270** can include a single pulse or more than two pulses. In a particular aspect, the protocol controller **140**, in response to detecting a particular count of pulses (e.g., two pulses) between the bus park **220** and the bus park **232**, determines that the read transaction is terminated unsuccessfully.

(50) A technical advantage of sending the termination pulses **270** is to limit a latency associated with a data read. For example, the read transaction is terminated if the data has not arrived at the subordinate device **122** prior to an end of the delay window **230**, and the data wire **134** is available for other commands.

(51) The added delay **262** of FIG. 2B has a first duration from a first time corresponding to an end of the bus park **220** to a second time corresponding to the protocol controller **142** sending the synchronization pulses **260** responsive to receiving the data **282** prior to the counter **246** reaching the value (e.g., 0) indicating no remaining clock cycles. The added delay **264** has a second duration from the first time to a third time corresponding to the protocol controller **142** sending the termination pulses **270** responsive to determining that the delay window **230** has ended (e.g., the counter **246** has reached the value indicating no remaining clock cycles) prior to arrival of the data at the subordinate device **122**. In a particular aspect, the second duration of the added delay **264** is longer than the first duration of the added delay **262**, and so a duration of the delay window **230** is longer than a duration of the delay window **228**.

(52) Although the protocol controller **142** is described with reference to FIG. 2A as sending the synchronization pulses **260** in response to detecting the arrival of the data **282** at the subordinate device **122** prior to the counter **246** reaching a particular value (e.g., 0), in some other implementations the protocol controller **142** can send the synchronization pulses **260** in response to the counter **246** reaching the particular value (e.g., 0) and detecting that the data **282** has arrived at the subordinate device **122**. In these implementations, the first duration of the added delay **262** can be equal to the second duration of the added delay **264**, the duration of the delay window **228** can be equal to the duration of the delay window **230**, and the duration of the added delays **262**, **264** and the duration of the delay windows **228**, **230** can be pre-determined based on the delay threshold **148**.

(53) Although the subordinate device **122** is described as using the counter **246** to track a duration of the added delay, in some other examples the principal device **120** can additionally, or in the alternative, track a duration of the added delay. For example, the protocol controller **140** initializes a counter concurrently with sending the register read command frame **202** based on a delay threshold **148** associated with the subordinate device **122**. For example, the counter is initialized based on the delay threshold **148** and a pre-determined extra delay. The protocol controller **140** updates the counter at each clock cycle of the clock **144** (e.g., a local clock). The subordinate device **122**, responsive to receiving the data **282** from the register **226**, sends the synchronization pulses **260** followed by the data **282**, as described with reference to FIG. 2A. The protocol controller **140**, in response to detecting the synchronization pulses **260** prior to the counter reaching

a value (e.g., 0) indicating no remaining clock cycles, reads the data **282** from the data wire **134** to complete the read transaction. Alternatively, the protocol controller **140**, in response to determining that counter has reached a value (e.g., 0) indicating no remaining clock cycles prior to detecting synchronization pulses **260** on the data wire **134**, terminates the read transaction.

(54) Referring to FIG. **3**, a particular implementation of a method **300** of configuration of a read synchronization delay at a subordinate device for 1 wire protocol is shown. In a particular aspect, one or more operations of the method **300** are performed by at least one of the protocol controller **142**, the subordinate device **122**, the device **102**, the system **100** of FIG. **1**, or a combination thereof.

(55) The method **300** includes, at block **302**, receiving, at a subordinate device on a data wire of a bus from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction. For example, the protocol controller **142** receives, at a subordinate device **122** on a data wire **134** of a serial bus **130** from a principal device **120**, a read command (e.g., the register read command frame **202** of FIG. **2A**) addressed to the subordinate device **122** indicating initiation of a read transaction, as described with reference to FIG. **2A**.

(56) The method **300** also includes, at block **304**, after a delay window, sending synchronization pulses from the subordinate device to the principal device based on a local clock of the subordinate device, where a duration of the delay window is greater than a clock cycle of the local clock. For example, the protocol controller **142** of the subordinate device **122**, after the delay window **228**, sends the synchronization pulses **260** from the subordinate device **122** to the principal device **120** based on the clock **146** (e.g., a local clock) of the subordinate device **122**, as described with reference to FIG. **2B**. A duration of the delay window **228** is greater than a clock cycle of the clock **146**.

(57) The method **300** further includes, at block **306**, after sending the synchronization pulses, sending data from the subordinate device on the data wire to the principal device based on the local clock during the read transaction. For example, the protocol controller **142** of the subordinate device **122**, after sending the synchronization pulses **260**, sends the data frame **204** representing the data **282** on the data wire **134** to the principal device **120** based on the clock **146** during the read transaction, as described with reference to FIG. **2B**.

(58) The method **300** enables the subordinate device **122** to send data that arrives subsequent to the end of the bus park **220** and prior to the end of the delay window **228**. The principal device **120** can thus retrieve valid data from the data wire **134** that has arrived at the subordinate device **122A** prior to the end of the delay window **228**.

(59) The method **300** of FIG. **3** may be implemented by a field-programmable gate array (FPGA) device, an application-specific integrated circuit (ASIC), a processing unit such as a central processing unit (CPU), a digital signal processor (DSP), a controller, another hardware device, firmware device, or any combination thereof. As an example, the method **300** of FIG. **3** may be performed by a processor that executes instructions, such as described with reference to FIG. **4**.

(60) FIG. **4** illustrates various electronic devices that may include or be integrated with the device **102**. For example, a mobile phone device **402**, a laptop computer device **404**, a fixed location terminal device **406**, a wearable device **408**, or a vehicle **410** (e.g., an automobile or an aerial device) may include a device **102** described herein. The devices **402**, **404**, **406** and **408** and the vehicle **410** illustrated in FIG. **4** are merely exemplary. Other electronic devices may also feature the device **102** including, but not limited to, a group of devices (e.g., electronic devices) that includes mobile devices, hand-held personal communication systems (PCS) units, portable data units such as personal digital assistants, global positioning system (GPS) enabled devices, navigation devices, set top boxes, music players, video players, entertainment units, fixed location data units such as meter reading equipment, communications devices, smartphones, tablet computers, computers, wearable devices (e.g., watches, glasses), Internet of things (IoT) devices, servers, routers, electronic devices implemented in vehicles (e.g., autonomous vehicles), or any

other device that stores or retrieves data or computer instructions, or any combination thereof.

(61) One or more of the components, processes, features, and/or functions illustrated in FIGS. **1-4** may be rearranged and/or combined into a single component, process, feature or function or embodied in several components, processes, or functions. Additional elements, components, processes, and/or functions may also be added without departing from the disclosure.

(62) In conjunction with the described implementations, an apparatus includes means for receiving, at a subordinate device on a data wire of a bus from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction. For example, the means for receiving can correspond to the protocol controller **142**, the subordinate device **122**, the data wire **134**, the device **102**, the system **100**, one or more other circuits or components configured to receive, at a subordinate device on a data wire of a bus from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction, or any combination thereof.

(63) The apparatus also includes means for sending, after a delay window, synchronization pulses from the subordinate device to the principal device based on a local clock of the subordinate device, where a duration of the delay window is greater than a clock cycle of the local clock. For example, the means for sending the synchronization pulses can correspond to the protocol controller **142**, the subordinate device **122**, the data wire **134**, the device **102**, the system **100**, one or more other circuits or components configured to send synchronization pulses after the delay window, or any combination thereof.

(64) The apparatus further includes means for sending data, after sending the synchronization pulses, from the subordinate device on the data wire to the principal device based on the local clock during the read transaction. For example, the means for sending the data can correspond to the protocol controller **142**, the subordinate device **122**, the data wire **134**, the device **102**, the system **100**, one or more other circuits or components configured to send data after sending the synchronization pulses, or any combination thereof.

(65) In some implementations, a non-transitory computer-readable medium (e.g., a computer-readable storage device, such as a memory) includes instructions that, when executed by one or more processors (e.g., the protocol controller **142**), cause the one or more processors to receive, at a subordinate device (e.g., a subordinate device **122**) on a data wire (e.g., the data wire **134**) of a bus (e.g., the serial bus **130**) from a principal device (e.g., the principal device **120**), a read command (e.g., the register read command frame **202**) addressed to the subordinate device indicating initiation of a read transaction. The instructions also cause the one or more processors to, after a delay window (e.g., the delay window **228**), send synchronization pulses (e.g., the synchronization pulses **260**) from the subordinate device to the principal device based on a local clock (e.g., the clock **146**) of the subordinate device, a duration of the delay window is greater than a clock cycle of the local clock. The instructions further cause the one or more processors to, after sending the synchronization pulses, send data (e.g., the data frame **204** representing the data **282**) from the subordinate device on the data wire to the principal device based on the local clock during the read transaction.

(66) Particular aspects of the disclosure are described below in sets of interrelated Examples: According to Example 1, a subordinate device includes a local clock; and a protocol controller coupled to a bus that includes a data wire, the protocol controller configured to: receive, on the data wire from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction; after a delay window, send synchronization pulses to the principal device based on the local clock, wherein a duration of the delay window is greater than a clock cycle of the local clock; and after sending the synchronization pulses, send data on the data wire to the principal device based on the local clock during the read transaction. Example 2 includes the subordinate device of Example 1, wherein the duration of the delay window is based on memory access latency, a configuration of the bus, or both. Example 3 includes the subordinate device of

Example 1 or Example 2, wherein the protocol controller is configured to, responsive to receiving the read command: initialize a counter based on the duration of the delay window; update the counter at each clock cycle of the local clock; and send the synchronization pulses in response to the counter reaching a particular value. Example 4 includes the subordinate device of any of Examples 1 to 3, wherein the duration of the delay window is pre-determined. Example 5 includes the subordinate device of any of Examples 1 to 4, wherein the duration of the delay window is based on a memory access time associated with arrival of the data. Example 6 includes the subordinate device of any of Examples 1 to 5, wherein the protocol controller is configured to initiate a memory access of the data responsive to receiving the read command; and send the synchronization pulses in response to arrival of the data. Example 7 includes the subordinate device of any of Examples 1 to 6, wherein the protocol controller is configured to, responsive to receiving the read command, initialize a counter and initiate a memory access of the data; and update the counter at each clock cycle of the local clock. Example 8 includes the subordinate device of Example 7, wherein the protocol controller is configured to send termination pulses to the principal device in response to the counter reaching a particular value prior to arrival of the data, the termination pulses corresponding to a termination of the read transaction. Example 9 includes the subordinate device of Example 7 or Example 8, wherein the protocol controller is configured to send the synchronization pulses to the principal device in response to arrival of the data prior to the counter reaching a particular value. According to Example 10, a method includes receiving, at a subordinate device on a data wire of a bus from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction; after a delay window, sending synchronization pulses from the subordinate device to the principal device based on a local clock of the subordinate device, wherein a duration of the delay window is greater than a clock cycle of the local clock; and after sending the synchronization pulses, sending data from the subordinate device on the data wire to the principal device based on the local clock during the read transaction. Example 11 includes the method of Example 10, wherein the duration of the delay window is based on memory access latency, a configuration of the bus, or both. Example 12 includes the method of Example 10 or Example 11, and further includes, responsive to receiving the read command: initializing a counter based on the duration of the delay window; and updating the counter at each clock cycle of the local clock, wherein the synchronization pulses are sent in response to the counter reaching a particular value. Example 13 includes the method of any of Examples 10 to 12, wherein the duration of the delay window is pre-determined. Example 14 includes the method of any of Examples 10 to 13, wherein the duration of the delay window is based on a memory access time associated with arrival of the data. Example 15 includes the method of any of Examples 10 to 14, and further includes initiating a memory access of the data responsive to receiving the read command, wherein the synchronization pulses are sent in response to arrival of the data. Example 16 includes the method of any of Examples 10 to 15, and further includes responsive to receiving the read command, initializing a counter and initiating a memory access of the data; and updating the counter at each clock cycle of the local clock. Example 17 includes the method of Example 16, wherein the synchronization pulses are sent to the principal device in response to arrival of the data prior to the counter reaching a particular value. According to Example 18, a non-transitory computer-readable medium stores instructions that, when executed by one or more processors, cause the one or more processors to receive, at a subordinate device on a data wire of a bus from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction; after a delay window, send synchronization pulses from the subordinate device to the principal device based on a local clock of the subordinate device, wherein a duration of the delay window is greater than a clock cycle of the local clock; and after sending the synchronization pulses, send data from the subordinate device on the data wire to the principal device based on the local clock during the read transaction. Example 19 includes the non-transitory computer-readable medium of Example 18, wherein the duration of the delay window is based on memory access

latency, a configuration of the bus, or both. Example 20 includes the non-transitory computer-readable medium of Example 18 or Example 19, wherein the instructions, when executed by the one or more processors, cause the one or more processors to, responsive to receiving the read command: initialize a counter based on the duration of the delay window; update the counter at each clock cycle of the local clock; and send the synchronization pulses in response to the counter reaching a particular value.

(67) Those of skill would further appreciate that the various illustrative logical blocks, configurations, modules, circuits, and algorithm steps described in connection with the implementations disclosed herein may be implemented as electronic hardware, computer software executed by a processor, or combinations of both. Various illustrative components, blocks, configurations, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or processor executable instructions depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application, such implementation decisions are not to be interpreted as causing a departure from the scope of the present disclosure.

(68) The steps of a method or algorithm described in connection with the implementations disclosed herein may be embodied directly in hardware, in a software module executed by a processor, or in a combination of the two. A software module may reside in random access memory (RAM), flash memory, read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), registers, hard disk, a removable disk, a compact disc read-only memory (CD-ROM), or any other form of non-transient storage medium known in the art. An exemplary storage medium is coupled to the processor such that the processor may read information from, and write information to, the storage medium. In the alternative, the storage medium may be integral to the processor. The processor and the storage medium may reside in an application-specific integrated circuit (ASIC). The ASIC may reside in a computing device or a user terminal. In the alternative, the processor and the storage medium may reside as discrete components in a computing device or user terminal.

(69) The previous description of the disclosed aspects is provided to enable a person skilled in the art to make or use the disclosed aspects. Various modifications to these aspects will be readily apparent to those skilled in the art, and the principles defined herein may be applied to other aspects without departing from the scope of the disclosure. Thus, the present disclosure is not intended to be limited to the aspects shown herein but is to be accorded the widest scope possible consistent with the principles and novel features as defined by the following claims.

Claims

1. A subordinate device comprising: a local clock; and a protocol controller coupled to a bus that includes a data wire, the protocol controller configured to: receive, on the data wire from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction; after a delay window, send synchronization pulses to the principal device based on the local clock, wherein a duration of the delay window is greater than a clock cycle of the local clock; and after sending the synchronization pulses, send data on the data wire to the principal device based on the local clock during the read transaction.
2. The subordinate device of claim 1, wherein the duration of the delay window is based on memory access latency, a configuration of the bus, or both.
3. The subordinate device of claim 1, wherein the protocol controller is configured to, responsive to receiving the read command: initialize a counter based on the duration of the delay window; update the counter at each clock cycle of the local clock; and send the synchronization pulses in response

to the counter reaching a particular value.

4. The subordinate device of claim 1, wherein the duration of the delay window is pre-determined.

5. The subordinate device of claim 1, wherein the duration of the delay window is based on a memory access time associated with arrival of the data.

6. The subordinate device of claim 1, wherein the protocol controller is configured to: initiate a memory access of the data responsive to receiving the read command; and send the synchronization pulses in response to arrival of the data.

7. The subordinate device of claim 1, wherein the protocol controller is configured to: responsive to receiving the read command, initialize a counter and initiate a memory access of the data; and update the counter at each clock cycle of the local clock.

8. The subordinate device of claim 7, wherein the protocol controller is configured to send termination pulses to the principal device in response to the counter reaching a particular value prior to arrival of the data, the termination pulses corresponding to a termination of the read transaction.

9. The subordinate device of claim 7, wherein the protocol controller is configured to send the synchronization pulses to the principal device in response to arrival of the data prior to the counter reaching a particular value.

10. A method comprising: receiving, at a subordinate device on a data wire of a bus from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction; after a delay window, sending synchronization pulses from the subordinate device to the principal device based on a local clock of the subordinate device, wherein a duration of the delay window is greater than a clock cycle of the local clock; and after sending the synchronization pulses, sending data from the subordinate device on the data wire to the principal device based on the local clock during the read transaction.

11. The method of claim 10, wherein the duration of the delay window is based on memory access latency, a configuration of the bus, or both.

12. The method of claim 10, further comprising, responsive to receiving the read command: initializing a counter based on the duration of the delay window; and updating the counter at each clock cycle of the local clock, wherein the synchronization pulses are sent in response to the counter reaching a particular value.

13. The method of claim 10, wherein the duration of the delay window is pre-determined.

14. The method of claim 10, wherein the duration of the delay window is based on a memory access time associated with arrival of the data.

15. The method of claim 10, further comprising initiating a memory access of the data responsive to receiving the read command, wherein the synchronization pulses are sent in response to arrival of the data.

16. The method of claim 10, further comprising: responsive to receiving the read command, initializing a counter and initiating a memory access of the data; and updating the counter at each clock cycle of the local clock.

17. The method of claim 16, wherein the synchronization pulses are sent to the principal device in response to arrival of the data prior to the counter reaching a particular value.

18. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to: receive, at a subordinate device on a data wire of a bus from a principal device, a read command addressed to the subordinate device indicating initiation of a read transaction; after a delay window, send synchronization pulses from the subordinate device to the principal device based on a local clock of the subordinate device, wherein a duration of the delay window is greater than a clock cycle of the local clock; and after sending the synchronization pulses, send data from the subordinate device on the data wire to the principal device based on the local clock during the read transaction.

19. The non-transitory computer-readable medium of claim 18, wherein the duration of the delay

window is based on memory access latency, a configuration of the bus, or both.

20. The non-transitory computer-readable medium of claim 18, wherein the instructions, when executed by the one or more processors, cause the one or more processors to, responsive to receiving the read command: initialize a counter based on the duration of the delay window; update the counter at each clock cycle of the local clock; and send the synchronization pulses in response to the counter reaching a particular value.
